

TARTU, Estonia

WMO: 262420

Lat: 58.2642N

Lon: 26.4614E

Elev: 72

StdP: 100.46

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.9	-18.2	-24.2	0.4	-21.6	-20.8	0.6	-17.8	9.6	-1.2	8.8	-0.6	1.6	80	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	27.9	19.6	26.0	18.5	24.2	17.6	20.8	25.7	19.6	24.2	18.5	22.9	3.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	23.2	17.9	13.0	21.9	16.8	12.1	20.7	60.2	25.6	56.1	24.2	52.5	22.9	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.6	6.8	DB	-25.0	30.5	4.6	1.6	-28.3	31.7	-31.0	32.6	-33.5	33.4	-36.8	34.6	(4)
(5)			WB	-25.2	22.6	4.5	1.2	-28.5	23.5	-31.1	24.2	-33.6	24.8	-36.9	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-4.9	-5.1	-0.9	5.9	11.2	15.2	18.1	16.7	12.0	6.2	1.6	-2.2
	DBStd	9.25	6.39	6.66	4.60	4.16	4.29	3.34	3.13	2.99	3.37	4.00	4.11	5.48
	HDD10.0	2171	463	422	337	135	37	2	0	0	19	128	251	378
	HDD18.3	4491	721	656	595	372	226	106	43	68	191	377	501	636
	CDD10.0	790	0	0	0	13	72	157	251	209	79	10	0	0
	CDD18.3	68	0	0	0	0	3	11	35	19	1	0	0	0
	CDH23.3	564	0	0	0	1	32	94	287	143	6	0	0	0
	CDH26.7	99	0	0	0	0	3	13	58	24	0	0	0	0
(14) Wind	WSAvg	3.2	3.4	3.2	3.4	3.3	3.0	3.0	2.7	2.7	2.9	3.2	3.5	3.7
(15) Precipitation	PrecAvg	612	35	27	29	34	51	64	75	81	65	55	52	42
	PrecMax	883	76	75	75	71	103	133	166	206	135	124	112	87
	PrecMin	394	5	7	4	1	9	6	9	15	16	10	11	13
	PrecStd	107	18	15	17	19	24	34	40	45	28	26	20	16
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	6.1	13.2	22.1	27.0	28.2	30.3	29.2	24.1	16.8	10.8	8.9
		MCWB	5.4	4.5	7.1	12.6	18.3	19.3	21.0	19.6	16.4	12.9	9.1	7.9
	2%	DB	4.4	4.6	9.1	18.9	23.9	26.1	28.3	26.8	21.0	14.3	9.1	6.8
		MCWB	3.6	3.4	5.1	10.9	15.8	18.1	20.1	19.1	15.4	12.0	7.9	5.9
	5%	DB	3.1	3.5	6.6	16.1	21.6	24.0	26.5	24.8	19.0	12.9	8.1	5.5
		MCWB	2.4	2.5	3.6	9.6	14.1	17.2	19.3	17.8	14.5	11.2	7.2	4.8
	10%	DB	2.1	2.4	4.8	13.4	19.2	21.9	24.7	22.9	17.3	11.7	7.0	4.1
		MCWB	1.5	1.6	2.7	8.1	12.9	15.8	18.4	17.0	13.7	10.2	6.3	3.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	4.8	7.7	13.7	19.6	20.9	22.6	21.7	17.7	13.7	9.5	7.8
		MCDB	6.4	5.5	12.2	19.7	24.9	26.1	27.8	26.4	21.1	15.5	10.2	8.7
	2%	WB	3.7	3.5	5.4	11.7	17.2	19.2	21.2	20.0	16.3	12.5	8.1	6.0
		MCDB	4.1	4.3	8.1	17.3	21.9	24.3	26.3	24.3	19.7	13.9	8.8	6.6
	5%	WB	2.4	2.6	4.1	10.1	15.3	17.9	20.1	18.7	15.2	11.5	7.3	4.7
		MCDB	2.9	3.3	6.0	15.0	19.8	22.5	24.9	23.0	18.1	12.8	7.9	5.3
	10%	WB	1.5	1.7	3.0	8.6	13.7	16.7	19.0	17.7	14.2	10.4	6.2	3.4
		MCDB	1.9	2.3	4.4	12.9	18.2	20.8	23.5	21.9	16.6	11.5	6.8	3.9
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.7	7.1	9.5	10.5	9.5	9.5	9.5	8.2	5.6	3.8	3.9
		MCDBR	3.6	4.3	9.0	14.0	13.7	12.1	12.2	12.3	10.4	6.6	4.5	3.7
		MCWBR	3.5	3.7	5.8	7.5	7.0	6.0	5.6	5.7	5.7	4.8	4.0	3.6
	5% WB	MCDBR	3.6	4.0	7.8	12.1	11.8	10.7	10.9	10.7	8.9	6.1	4.1	3.4
		MCWBR	3.6	3.7	5.5	6.9	6.8	6.0	5.5	5.7	5.4	4.8	3.9	3.5
(40) Clear-Sky Solar Irradiance	taub		0.274	0.305	0.335	0.368	0.363	0.362	0.387	0.384	0.354	0.334	0.312	0.267
	taud		2.123	2.137	2.195	2.320	2.395	2.424	2.374	2.379	2.449	2.401	2.219	2.111
	Ebn at Noon		546	707	800	839	870	875	841	815	786	686	496	427
	Edh at Noon		51	81	102	108	107	105	109	101	81	64	48	38
(44) All-Sky Solar Radiation	RadAvg		0.36	1.02	2.47	3.89	5.29	5.41	5.22	4.13	2.63	1.16	0.36	0.22
	RadStd		0.07	0.18	0.31	0.47	0.56	0.45	0.53	0.42	0.31	0.17	0.05	0.04

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	-163	-165	N/A	N/A

Nomenclature: See separate page